

# Final Project Report

## Precision Horticulture An Automated Planter System Utilizing Soil Moisture Sensors

(NOTE #1) THE PLANTER SYSTEMS FRAME IS CONSTRUCTED FROM WEATHER-TREATED LUMBER, WHICH WAS THEN STAINED TO ACHIEVE THE DESIRED COLOR AND SEALED TO PROVIDE LONG TERM PROTECTION AGAINST THE ELEMENTS. SWIVEL CASTER WHEELS ARE INSTALLED ON THE BASE OF EACH LEG, MAKING IT EASY TO MOVE THE SYSTEM DESPITE ITS SIGNIFICANT WEIGHT—IT TAKES TWO PEOPLE TO LIFT SAFELY. THE OVERALL HEIGHT WAS CAREFULLY SET TO JUST BELOW CHEST LEVEL, ENSURING COMFORTABLE, ERGONOMIC ACCESS FOR PLANTING AND MAINTENANCE.

FIVE-INCH DECK SCREWS WERE USED TO SECURE THE LEGS IN PLACE, DESIGNED SO THAT THE MOUNTING HARDWARE DOES NOT INTERFERE WITH, OR REDUCE, THE AVAILABLE PLANTING SPACE. THIS RESULTS IN CLEAN, SQUARE CORNERS THAT NOT ONLY MAXIMIZE THE USABLE VOLUME BUT ALSO PROVIDE A SIMPLE PATHWAY FOR INTEGRATING A DRIP IRRIGATION Lining IN THE FUTURE. THIS DRIP Lining COULD FEED INTO A FILTRATION AND RECYCLING SYSTEM, ENABLING RUNOFF WATER TO BE ROUTED BACK TO THE STORAGE DRUM FOR SUSTAINABLE REUSE.



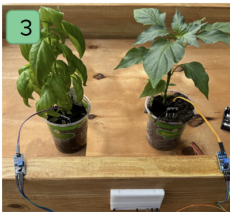
1 2 3 4 5 5R 6

(NOTE #2) THIS VIEW HIGHLIGHTS THE "PRESENTATION" SIDE OF THE PLANTER SYSTEM, INTENTIONALLY DESIGNED TO CONCEAL ALL WIRING FOR A CLEAN, SEAMLESS APPEARANCE. NONE OF THE ELECTRICAL COMPONENTS ARE VISIBLE, RESULTING IN A POLISHED LOOK FOR THE PROJECT. YOU'LL ALSO NOTICE THAT THE MOTORS FOR BOTH SOIL ZONES, AS WELL AS THE MANUAL DISPENSING PUMPS, ARE NOT POSITIONED INSIDE THE WATER DRUM IN THIS IMAGE. THIS IS BECAUSE THE MANUFACTURER-SUPPLIED HOSES WERE NOT LONG ENOUGH TO EFFECTIVELY IRRIGATE THE SOIL ZONES AS INTENDED. AFTER EXPERIMENTING WITH SEVERAL CONFIGURATIONS, IT WAS CLEAR THAT THE STANDARD HOSE LENGTHS WERE INSUFFICIENT. LONGER REPLACEMENT HOSES HAVE BEEN ORDERED; THESE WILL BE FEATURED IN THE UPCOMING VIDEO DEMONSTRATION, ONCE THE SYSTEM IS FULLY ASSEMBLED AND OPTIMIZED.



2

(NOTE #3) EACH LM3933 SOIL MOISTURE SENSOR ENABLES REAL-TIME SOIL DIAGNOSTICS AND PROMPTED IRRIGATION CYCLES WHENEVER MOISTURE LEVELS DROPPED BELOW RESEARCHED SOIL THRESHOLDS. EACH PLANT ZONE USED A DEDICATED 5V WATER PUMP, CONTROLLED BY RELAY MODULES WITH DIGITAL OUTPUTS, TO DELIVER PRECISE WATER DOSES AND MINIMIZE THE RISK OF OVER- OR UNDER-WATERING.



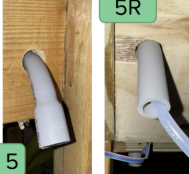
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(NOTE #4) THE ROCK SHELF WAS ORIGINALLY MEANT TO HOLD A FIVE-GALLON WATER DRUM, BUT WAS LATER RAISED TO STORE PLANTS, SOIL, AND TOOLS OFF THE GROUND. ITS WEIGHT CAPACITY WAS ALSO INCREASED TO SUPPORT A FUTURE TWENTY-GALLON WATER DRUM, REDUCING HOW OFTEN IT NEEDS TO BE REFILLED.



4

(NOTES #5 & 5R) TO ENHANCE FLEXIBILITY, I INCORPORATED A PROXIMITY-ACTIVATED MANUAL REFILL FEATURE USING AN HC-SR04 ULTRASONIC SENSOR, WHICH DETECTS WHEN A HAND IS WITHIN 5 CENTIMETERS. ONCE TRIGGERED, THE SIDE HOSE DISPENSES WATER AFTER A BRIEF DELAY AND CONTINUES TO FLOW AS LONG AS HAND PRESENCE IS DETECTED, PROVIDING CONVENIENT SUPPLEMENTAL WATERING WHENEVER NEEDED.



5 5R

(NOTE #6) FOR CONTROL AND SAFETY, I INSTALLED A SET OF THREE SPST ROCKER SWITCHES WIRED WITH INPUT PULL-UP, ALLOWING EACH SOIL ZONE AND THE MANUAL DISPENSING STATION TO BE INDEPENDENTLY ENABLED OR DISABLED. THIS SETUP MAKES IT EASY TO TURN OFF THE SENSORS AND PUMPS DURING MAINTENANCE, OR WHEN REMOVING SENSORS, EFFECTIVELY PREVENTING ACCIDENTAL WATERING THAT COULD HAPPEN IF DISCONNECTED OR MOVED PROBES ARE MISREAD AS "DRY."



6



## Project Summary:

### Precision Horticulture: An Automated Planter System Utilizing Soil Moisture Sensors

In this project, I explored the design and implementation of an automated planter system that utilized soil moisture sensors to facilitate “precision horticulture” at an attainable price range. My objective was to construct a device capable of independently monitoring two regions and regulating soil moisture, complemented by a water refill station to allow for manual water dispensing activated by an infrared sensor.

I built the system on an Arduino microcontroller, which managed two LM393 soil moisture sensors connected to analog pins (A0, A1). These sensors enabled real-time soil diagnostics and prompted irrigation cycles whenever moisture levels dropped below researched soil thresholds (400 for saturated soil, 550 as the trigger for watering). Each plant zone used a dedicated 5V water pump, controlled by relay modules with digital outputs, to deliver precise water doses and minimize the risk of over- or under-watering.

To add flexibility, I included a proximity-activated manual refill component employing an HC-SR04 ultrasonic sensor to detect hand proximity within 5 centimeters. When activated, the side hose dispenses water after a short delay and maintains flow as long as presence was detected, allowing for supplemental watering as needed.

For control and safety, I included a bundle of (3) SPST rocker switches (wired using INPUT\_PULLUP) to independently enable or disable circuits for each soil zone and the manual dispensing station. This approach ensured that soil sensors and pumps were disabled during maintenance or when sensors were removed, effectively preventing unwanted watering that could occur if removed or moving probes registered as “dry.”

In the future I plan to include power management using a low-power operational mode via the LowPower library. When I engage sleep mode through its dedicated switch, the system will enter deep-sleep states for 60-minute intervals between sensor readings, significantly extending battery-powered operating life.

Throughout development, I faced technical challenges in calibrating soil moisture thresholds for consistent plant health and in managing sensor removal scenarios with the safety switch logic. The resulting automated planter system demonstrated reliable, energy-efficient, and modular precision irrigation capabilities. I found that this setup met my goals for smart horticulture while providing a scalable foundation for future enhancements in environmental monitoring and control.

See code below.

## Arduino Code:

```
int soilPin1  = A0;
int relayPin1 = 6;
int MOIST_DRY1 = 600;
int switchPin1 = 2; // Rocker switch for Pump 1

int soilPin2  = A1;
int relayPin2 = 7;
int MOIST_DRY2 = 600;
int switchPin2 = 3; // Rocker switch for Pump 2

int pumpTime  = 8000;

int irPin     = 9; // IR sensor OUT for Pump 3
int relayPin3 = 8; // Relay for Pump 3
int switchPin3 = 4; // Rocker switch for Pump 3

void setup() {
  pinMode(relayPin1, OUTPUT); digitalWrite(relayPin1, HIGH);
  pinMode(relayPin2, OUTPUT); digitalWrite(relayPin2, HIGH);
  pinMode(relayPin3, OUTPUT); digitalWrite(relayPin3, HIGH);

  pinMode(switchPin1, INPUT_PULLUP);
  pinMode(switchPin2, INPUT_PULLUP);
  pinMode(switchPin3, INPUT_PULLUP);
  pinMode(irPin, INPUT);

  Serial.begin(9600);
}

void loop() {
  // Pump 1 (Soil)
  int reading1 = analogRead(soilPin1);
  if (digitalRead(switchPin1) == LOW) { // Switch ON (closed)
    Serial.print("Pump 1 Switch: ON | ");
    Serial.print("Soil sensor 1 reading: ");
    Serial.println(reading1);
    digitalWrite(relayPin1, HIGH);
    if (reading1 >= MOIST_DRY1) {
      Serial.println("Soil 1 is DRY — Pump 1 ON!");
      digitalWrite(relayPin1, LOW);
      delay(pumpTime);
      digitalWrite(relayPin1, HIGH);
    }
  }
}
```

```
    Serial.println("Pump 1 OFF\n");
  } else {
    Serial.println("Soil 1 moist — Pump 1 OFF\n");
  }
} else {
  digitalWrite(relayPin1, HIGH); // Always ensure OFF
  Serial.println("Pump 1 Switch: OFF | Pump 1 OFF\n");
}

// Pump 2 (Soil)
int reading2 = analogRead(soilPin2);
if (digitalRead(switchPin2) == LOW) { // Switch ON (closed)
  Serial.print("Pump 2 Switch: ON | ");
  Serial.print("Soil sensor 2 reading: ");
  Serial.println(reading2);
  digitalWrite(relayPin2, HIGH);
  if (reading2 >= MOIST_DRY2) {
    Serial.println("Soil 2 is DRY — Pump 2 ON!");
    digitalWrite(relayPin2, LOW);
    delay(pumpTime);
    digitalWrite(relayPin2, HIGH);
    Serial.println("Pump 2 OFF\n");
  } else {
    Serial.println("Soil 2 moist — Pump 2 OFF\n");
  }
} else {
  digitalWrite(relayPin2, HIGH);
  Serial.println("Pump 2 Switch: OFF | Pump 2 OFF\n");
}

// Pump 3 (IR/manual)
int irValue = digitalRead(irPin); // LOW = hand/object detected
if (digitalRead(switchPin3) == LOW) { // Switch ON
  Serial.print("Pump 3 Switch: ON | IR sensor: ");
  Serial.println(irValue == HIGH ? "Object detected" : "No object");
  if (irValue == HIGH) {
    Serial.println("Hand/object detected by IR — Pump 3 ON!");
    digitalWrite(relayPin3, LOW);
    delay(pumpTime);
    digitalWrite(relayPin3, HIGH);
    Serial.println("Pump 3 OFF\n");
  } else {
    digitalWrite(relayPin3, HIGH);
    Serial.println("No object detected — Pump 3 OFF\n");
  }
}
```



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```
    }  
  } else {  
    digitalWrite(relayPin3, HIGH);  
    Serial.println("Pump 3 Switch: OFF | Pump 3 OFF\n");  
  }  
  
  Serial.println("-----");  
  delay(2000);  
}
```

## Circuit Diagram:

